

Lucía Cabrera

luciacabrera.ar



Pick a flower on Earth and
you move the farthest star.
– Paul Dirac

Areas of specialization

• Theoretical & Mathematical
Physics • String Theory &
M-Theory • AI Evaluation &
Benchmarking

Languages

Spanish	C2	mother tongue
English	C2	• • • • •
German	B2	• • • • •
Italian	B2	• • • • •

Programming Languages

L ^A T _E X	• • • • •
haskell	• • • • •
python	• • • • •
lean	• • • • •

Dev Stack

Git	• • • • •
Jupyter Notebooks	• • • • •
OpenAI API	• • • • •

Other Interests

• Functional Programming
• Quantitative Research
• Structured AI Reasoning
• Renaissance Music
• Philosophy



EDUCATION

2021 – exp. 2026

Physics Ph.D. Student

BALSEIRO INSTITUTE · Bariloche, Argentina

Thesis: Particle Physics, Quantum Gravity and Dualities in String Theory.

Supervisor: Prof. Gerardo Aldazabal **Co-supervisor:** Prof. Bobby Acharya



2019

Physics Licentiate Degree

NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina

Thesis: Dual Holographic Description of Yang-Mills Plasmas **Supervisor:** Prof. Martin Schvellinger (La Plata Institute of Physics - National University of La Plata). **GPA:** 9.53/10



EXPERIENCE

2025 –

AI Evaluator & Benchmarking Researcher

POINDEXTER LABS · London, United Kingdom

• Designed adversarial math/physics datasets to probe LLM reasoning
• Contributed to AI safety prompting (Forum AI collab) • Analyzed model failure modes & co-authored methodology paper on prompt sensitivity.



2022 – 2025

Teaching Assistant

BALSEIRO INSTITUTE · Bariloche, Argentina

• Instructed undergraduate courses in linear algebra, differential equations, and complex analysis.



2016 – 2019

Teaching Assistant

NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina

• Instructed undergraduate courses in linear/spectral algebra, complex analysis and distribution theory.



MANUSCRIPTS

2026

Too long; didn't solve · Accepted to EvalEval Workshop @ ACL 2026

arXiv:2604.07593 [cs.AI]

L. CABRERA, I. SAXTON-KNIGHT, J. D'ARCY

On how prompt length shapes LLM behaviour and discriminativeness.

In preparation

On Capturing Stringy Features within a Field Theory Description

G. ALDAZABAL, E. ANDRÉS, L. CABRERA, M. MAYO

Embedding stringy features into a QFT and its hidden symmetries.

In preparation

Branches of M-Theory/Heterotic string theory duals

B. ACHARYA, D. BALDWIN, L. CABRERA, K. NARAIN

A dictionary of heterotic string theories dual to M-theory on G_2 manifolds.

HIGHLIGHTED SCHOLARSHIPS & AWARDS

2025

ZONTA Women in STEM Award – South-American Winner

ZONTA INTERNATIONAL



2023

STEP Fellowship

INTERNATIONAL CENTRE FOR THEORETICAL PHYSICS · Trieste, Italy

Tri-annual international research collaboration fellowship.



2020

Highest GPA in Physics

PUNTO BIZ MAGAZINE · Santa Fe, Argentina

Highest GPA in a physics degree in the province of Santa Fe, Argentina.



2019

Doctoral Fellowship

NATIONAL SCIENTIFIC AND TECHNICAL RESEARCH COUNCIL · Argentina



2019

Summer Research Scholarship

BALSEIRO INSTITUTE · Bariloche, Argentina

Research internship. **Project:** Communication Servers for LAHN/ANDES.



INVITED TALKS

2024

Invited Talk · Imperial College London. United Kingdom.

2024

Contributed Talk · 5th Young Frontiers Meeting. Online.

2024

Contributed Talk · 1st Online Conference for Women in Theoretical Physics from LATAM. Online.

RESEARCH VISITS

2025

ICTP – as STEP fellow. Italy.

2024

ICTP – as STEP fellow. Italy.

2023

ICTP – as STEP fellow. Italy.

2022

IPhT Saclay. France.